

W5 – Variational Inference (Part I)

Instructor: Edwin V. Bonilla

School of Computer Science and Engineering

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Acknowledgements

Material derived from:

- [\[Bishop, PRML, 2006\]](#) Pattern Recognition and Machine Learning, Christopher Bishop, 2006
 - Almost all figures in today's lecture are from this book
- [\[Murphy, MLaPP, 2012\]](#) Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, 2012

Aims

This lecture will allow you to understand variational inference methods for posterior estimation in graphical models. Following it you should be able to:

- Understand the expectation-maximisation (EM) algorithm from a more general perspective, identifying the objective function it is maximising and the relations between its constituents
- Understand and apply properties of factorised distributions for posterior approximation
- Exploit conjugacy properties for obtaining closed-form updates in variational algorithms
- Identify advantages and disadvantages of variational methods for approximate Bayesian inference

Outline

- I. EM Revisited
- II. Variational Inference
- III. Factorised Distributions
- IV. Bayesian GMMs
- V. Remarks and Conclusions

I. EM Revisited

The EM Algorithm Revisited (1)

$\mathbf{x}^{(i)}$: Observed variable for data point $i \rightarrow \mathbf{X} = \{\mathbf{x}^{(i)}\}$

$\mathbf{z}^{(i)}$: Hidden or missing variable $\rightarrow \mathbf{Z} = \{\mathbf{z}^{(i)}\}$

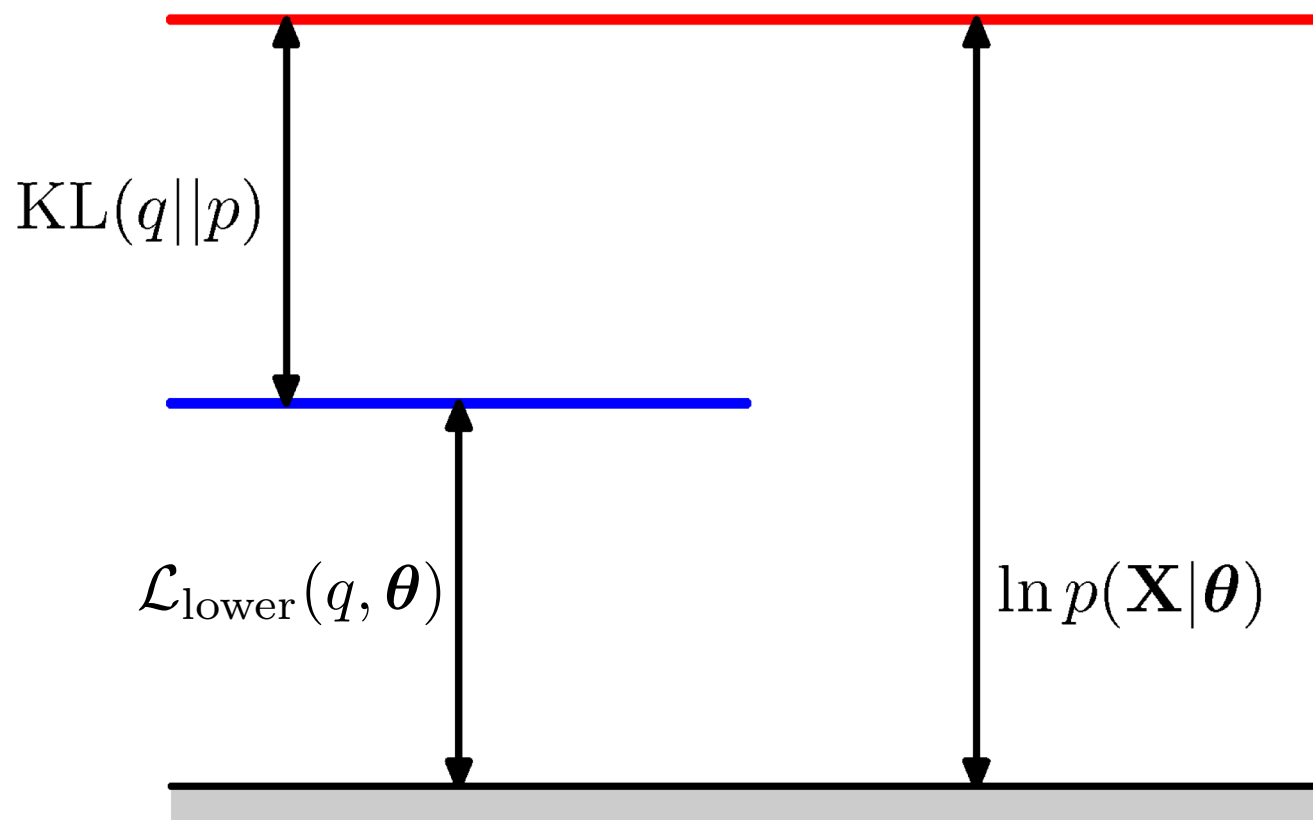
- Direct maximisation of the data log likelihood $\log p(\mathbf{X}|\boldsymbol{\theta})$ is hard
 - Optimization of the expected complete data log likelihood is significantly easier
- We can exploit the decomposition:

$$\log p(\mathbf{X}|\boldsymbol{\theta}) = \text{KL}(q(\mathbf{Z}|\mathbf{X})||p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta})) + \mathcal{L}_{\text{lower}}(q, \boldsymbol{\theta})$$

- Where: $\mathcal{L}_{\text{lower}}(q, \boldsymbol{\theta}) \stackrel{\text{def}}{=} \mathbb{E}_{q(\mathbf{Z}|\mathbf{X})} \left[\log \frac{p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})}{q(\mathbf{Z}|\mathbf{X})} \right]$
 - Is a lower bound on the (marginal) data log likelihood

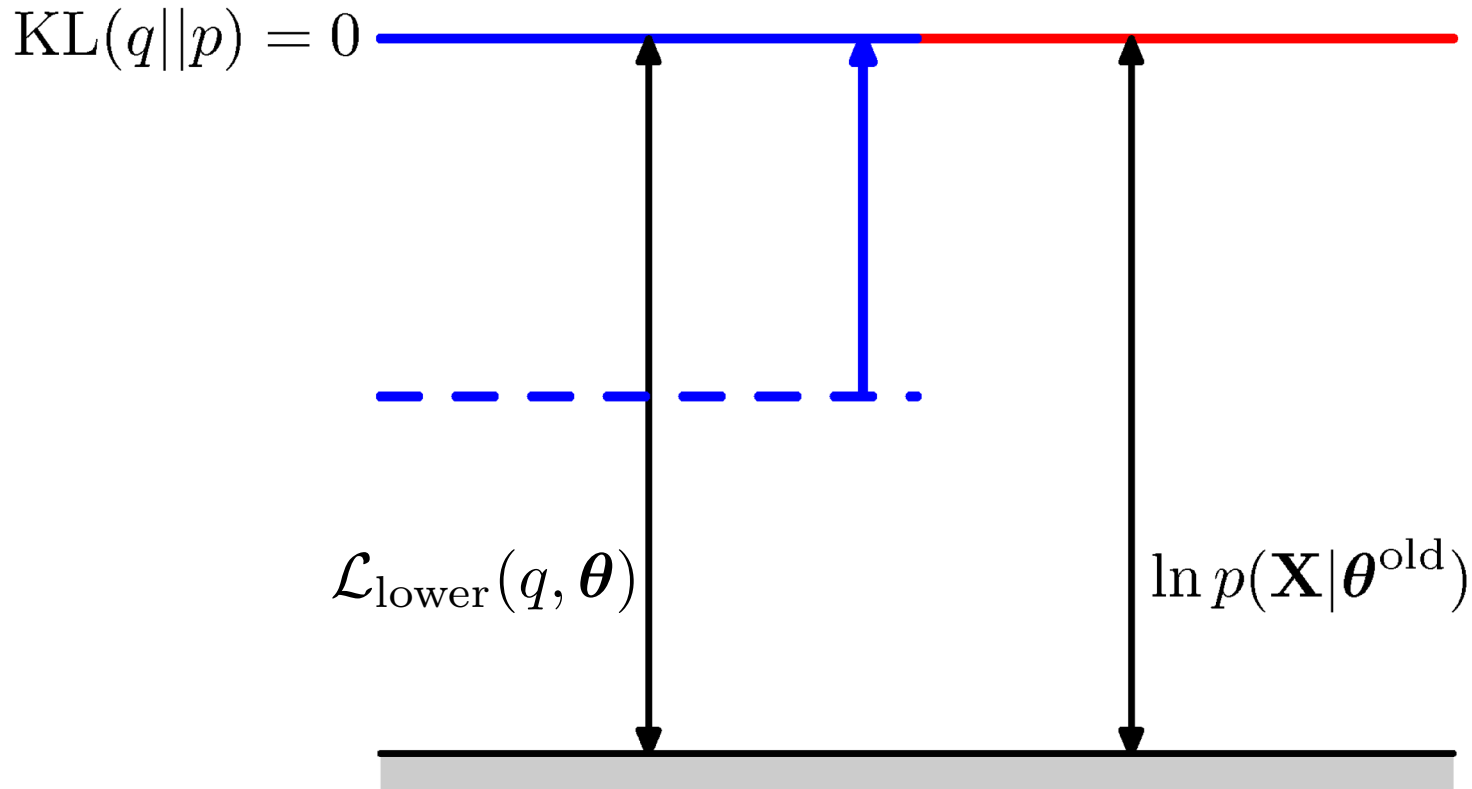
The EM Algorithm Revisited (2)

Decomposition of the marginal likelihood



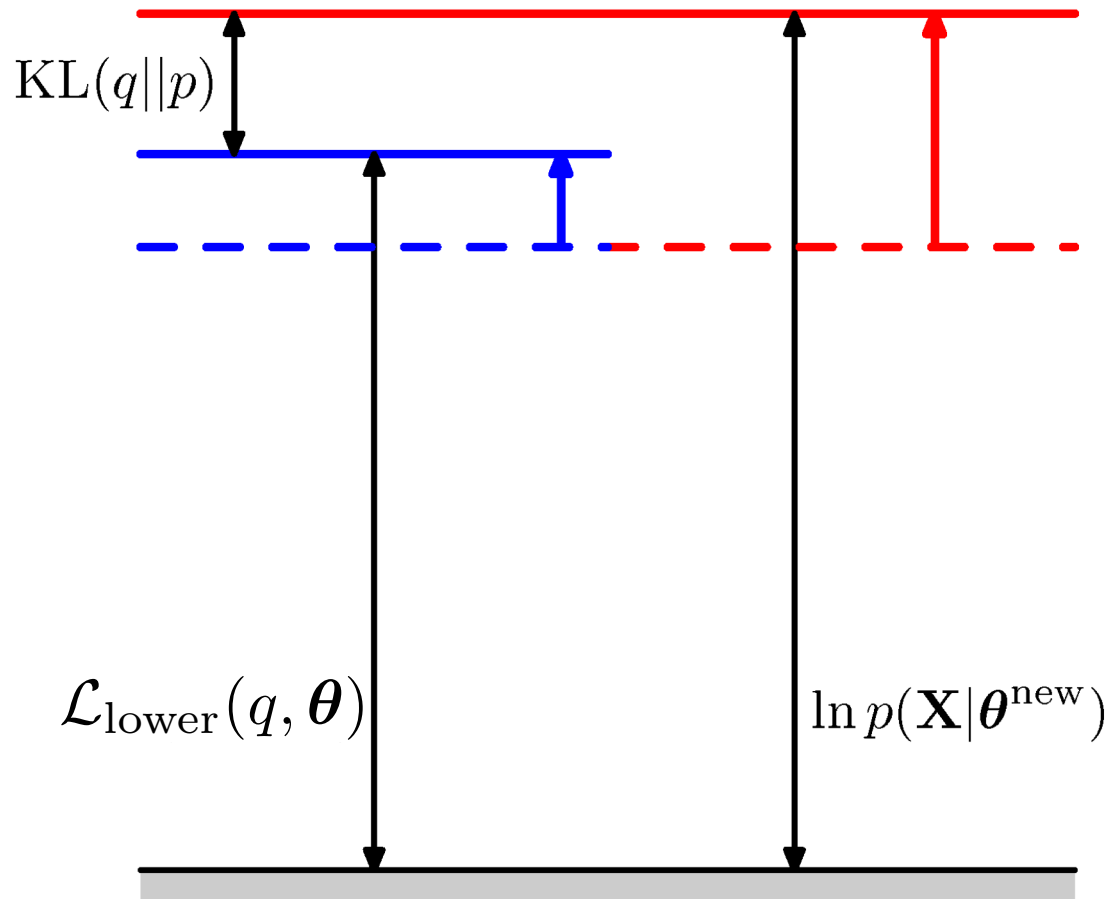
The EM Algorithm Revisited (3)

- E-step: Maximisation of $\mathcal{L}_{\text{lower}}(q, \theta)$ wrt $q(\mathbf{Z}|\mathbf{X})$
 - Optimal setting is the true posterior $q(\mathbf{Z}|\mathbf{X}) = p(\mathbf{Z}|\mathbf{X}, \theta)$



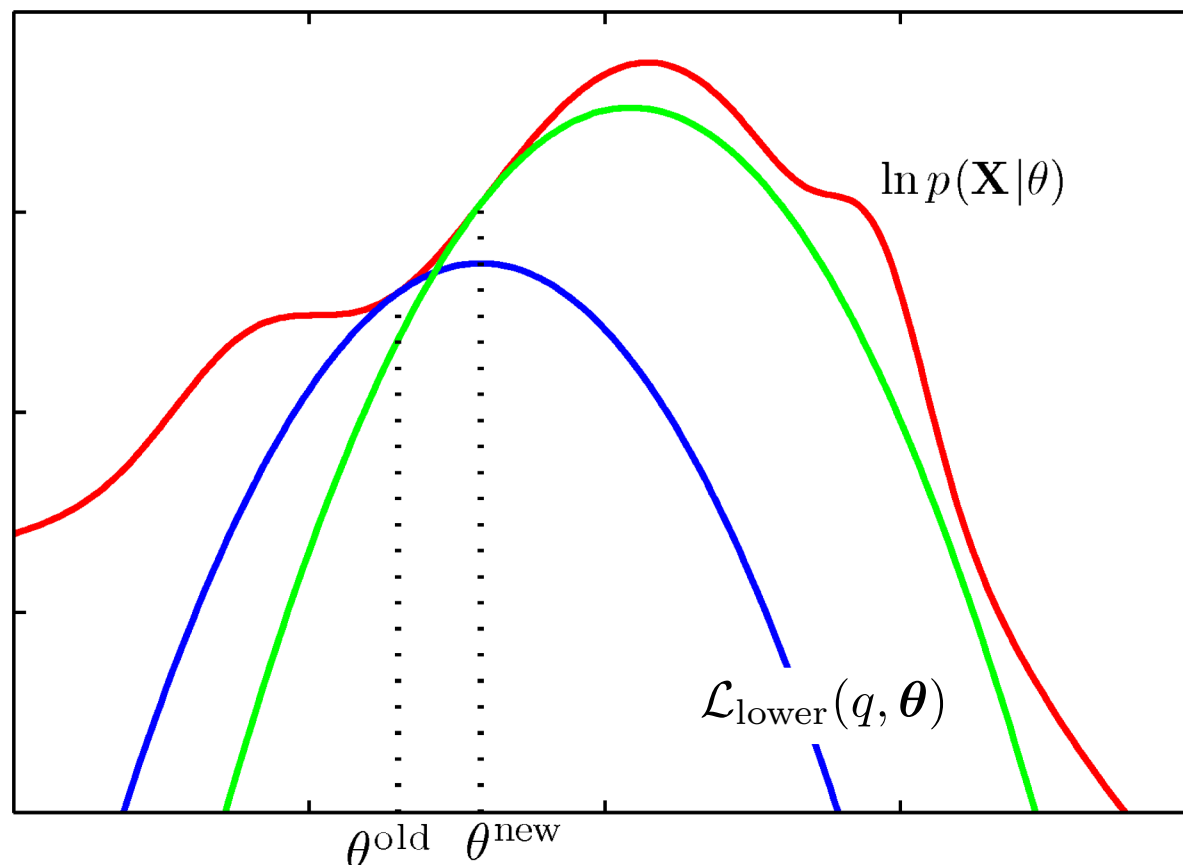
The EM Algorithm Revisited (4)

- M-step: Maximisation of $\mathcal{L}_{\text{lower}}(q, \theta)$ wrt θ
 - Since $q(\mathbf{Z}|\mathbf{X})$ is fixed, moving θ also increases the KL
 - Objective optimised is in fact $Q(\theta, \theta^{\text{old}})$ as $\mathcal{L}_{\text{lower}}(q, \theta) = Q(\theta, \theta^{\text{old}}) + \boxed{\mathbb{H}(q)}$



The EM Algorithm Revisited (5)

- EM in parameter space
 - [E] Updating $q \rightarrow$ tight bound (blue curve touches red curve)
 - [M] Updating $\theta \rightarrow$ leaves a gap (max of blue curve at θ^{new})
 - [E] green curve



Generalisations of the EM Algorithm

- **Vanilla EM** indirectly maximises the likelihood
 - E-step and M-step are sometimes simple (e.g. GMMs)
 - However, they may be intractable for complex models
- **Generalized EM**
 - Deals with intractable M-step by partially optimizing $\mathcal{L}_{\text{lower}}(q, \theta)$ wrt θ
 - » As before, each cycle will improve the likelihood (not get worse)
 - » Can use standard gradient-based optimization techniques
 - » Expectation conditional maximisation
 - Alternating optimisation of parameter subsets
- **Intractable E-step**
 - Similarly, we can partially optimize $\mathcal{L}_{\text{lower}}(q, \theta)$ wrt $q(\mathbf{Z}|\mathbf{X})$
 - » Any local maximum of $\mathcal{L}_{\text{lower}}(q, \theta)$ will also be a local maximum $\log p(\mathbf{X}|\theta)$

Such generalizations naturally lead us to variational inference

The Need for Approximate Inference

- Computation of posterior (or expectations over it) is crucial:

$$p(\mathbf{Z}|\mathbf{X}) = \frac{\overset{\text{Likelihood}}{p(\mathbf{X}|\mathbf{Z})} \overset{\text{prior}}{p(\mathbf{Z})}}{\underset{\text{marginal}}{p(\mathbf{X})}}$$

- Exact inference through conjugate priors for simple models
- JTA in discrete case $\rightarrow$ exponential in tree width
- EM requires expectations over posterior for parameter estimation
- For many models exact computation is unfeasible
 - » Intractability of marginal likelihood (sums or integrals in high dimensions)

Stochastic approaches

- Sampling
- Exact results given infinite computation
- Usually demanding, non-scalable

Deterministic approaches

- Assumptions on (approximate posterior)
- Almost never exact
- Usually scalable
- *Variational inference*

II. Variational Inference

Variational Inference and Calculus of Variations

Standard calculus

- Function
 - Input: variable
 - Output: Scalar

- Derivatives of functions
 - Changes in the output wrt infinitesimal changes in the input

Calculus of variations

- Functionals
 - Input: Function
 - Output: scalar

$$\mathbb{H}[q(\mathbf{x})] = - \int p(\mathbf{x}) \log p(\mathbf{x}) d\mathbf{x}$$

- Functional derivatives
 - Changes in the output wrt infinitesimal changes in the input
 - » Similar rules to std calculus

Main idea: Approximate inference as an optimisation problem

- Objective is a functional (input is the posterior distribution)
 - » Constrain the posterior to a suitable family of functions
 - » Optimise wrt (approximate) posterior

The Variational Objective

$\mathbf{X}$: Observed variables

$\mathbf{Z}$: Hidden or missing variables

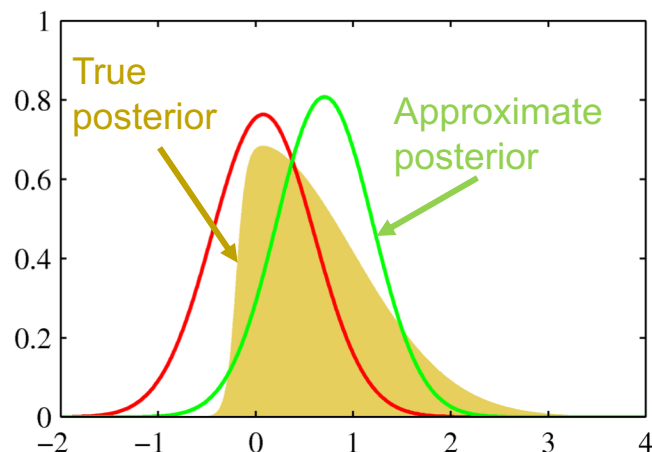
- **Goal:** given prior $P(\mathbf{Z})$ and conditional likelihood $p(\mathbf{X}|\mathbf{Z}) \rightarrow$ approximate the posterior $p(\mathbf{Z}|\mathbf{X})$ with $q(\mathbf{Z}|\mathbf{X})$
 - Omitting θ as we can include them in $\mathbf{Z}$ as random variables
- We have seen that

$$\log p(\mathbf{X}) = \text{KL}(\overset{\text{Approximate}}{q(\mathbf{Z}|\mathbf{X})} \parallel \overset{\text{True}}{p(\mathbf{Z}|\mathbf{X})}) + \mathcal{L}_{\text{lower}}(q)$$

- Where: $\mathcal{L}_{\text{lower}}(q) \stackrel{\text{def}}{=} \mathbb{E}_{q(\mathbf{Z}|\mathbf{X})} \left[\log \frac{p(\mathbf{X}, \mathbf{Z})}{q(\mathbf{Z}|\mathbf{X})} \right]$ This is our variational objective (functional)
 - We will attempt to maximise $\mathcal{L}_{\text{lower}}(q)$ wrt $q(\mathbf{Z}|\mathbf{X})$
 - This is indeed equivalent to minimizing $\text{KL}(q(\mathbf{Z}|\mathbf{X}) \parallel p(\mathbf{Z}|\mathbf{X}))$

What exactly is $q(\mathbf{Z}|\mathbf{X})$?

- Free-form $q(\mathbf{Z}|\mathbf{X})$: optimisation of the functional $\mathcal{L}_{\text{lower}}(q)$
 - Would give us the right answer as the KL vanishes at the true posterior
 - However, need to solve normalization, which was our initial problem!
- Fixed-form $q(\mathbf{Z}|\mathbf{X})$: Consider a restricted family of distributions
 - Minimize the objective wrt members of this family
 - E.g. Use factorised distribution
 - E.g. Use a parametrized distribution $q(\mathbf{Z}|\mathbf{X},\lambda)$
 - » Optimisation via standard calculus



What family of distributions?

- As flexible as possible
- Tractability is the main constraint
- No risk of overfitting
 - The more flexible the better the approximation to the true posterior

Understanding the Variational Objective

The lower bound $\mathcal{L}_{\text{lower}}(q) \stackrel{\text{def}}{=} \mathbb{E}_{q(\mathbf{Z}|\mathbf{X})} \left[\log \frac{p(\mathbf{X}, \mathbf{Z})}{q(\mathbf{Z}|\mathbf{X})} \right]$ can be written as:

$$\mathcal{L}_{\text{lower}}(q) = \underbrace{\mathbb{E}_{q(\mathbf{Z}|\mathbf{X})} [\log p(\mathbf{X}|\mathbf{Z})]}_{\text{Expected log likelihood (ELL)}} - \underbrace{\text{KL}(q(\mathbf{Z}|\mathbf{X}) || p(\mathbf{Z}))}_{\text{KL (approx. posterior || prior)}}$$

- **ELL term is a model fit:** How well the (samples from the) posterior explains the observations
- **KL is a penalty term:** Keep posterior close to prior beliefs
- Also known as
 - Variational free energy
 - Evidence lower bound (ELBO)

III. Factorised Distributions

Factorised Distributions

Mean field approximation

- For notational simplicity, make $q(\mathbf{Z}) \stackrel{\text{def}}{=} q(\mathbf{Z}|\mathbf{X})$ and assume that it factorises over M disjoint groups:

$$q(\mathbf{Z}) = \prod_{i=1}^M q_i(\mathbf{Z}_i)$$

- In this case, we can optimise the variational objective in “free-form”
 - No additional assumptions on the functional forms of $q_i(\mathbf{Z}_i)$

Optimal
solution

$$\log q_j^*(\mathbf{Z}_j) = \boxed{\mathbb{E}_{i \neq j} [\log p(\mathbf{X}, \mathbf{Z})]} + \boxed{\text{const.}}$$

Expectation over all $q(\mathbf{Z}_i)$
except $\mathbf{Z}_j$ of the log joint

Additive constant set by
normalisation

- General consistency conditions
 - » Need to iterate
 - » Guaranteed convergence as bound is convex wrt to each factor

Properties of Factorised Distributions (1)

- Consider a 2-dimensional Gaussian $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})$

$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \quad \boldsymbol{\Lambda} = \begin{bmatrix} \Lambda_{11} & \Lambda_{12} \\ \Lambda_{21} & \Lambda_{22} \end{bmatrix} \quad \Lambda_{21} = \Lambda_{12}$$

Precision matrix

- Goal:** Approximate $p(\mathbf{z})$ with a $q(\mathbf{z}) = q_1(z_1) q_2(z_2)$
 - No assumptions on the functional form of the approximating distributions

- Using the general mean-field update we obtain:

$$\log q_1^*(z_1) = \mathbb{E}_{z_2} [\log p(\mathbf{z})] + \text{const.}$$

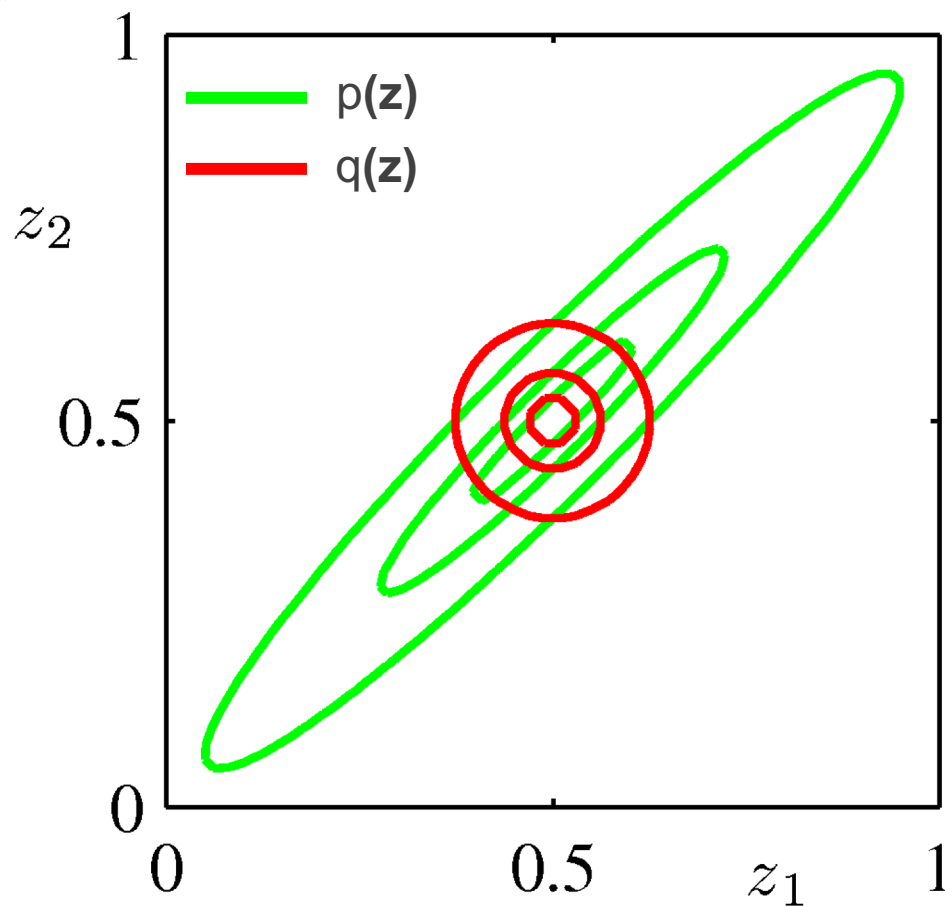
– and similarly for q_2

- In general, we need to iterate but here we have a closed-form solution

$$q_i^*(z_i) = \mathcal{N}(z_i | \mu_i, \Lambda_{ii}^{-1})$$

- Correct mean
- Variance?

Properties of Factorised Distributions (2)



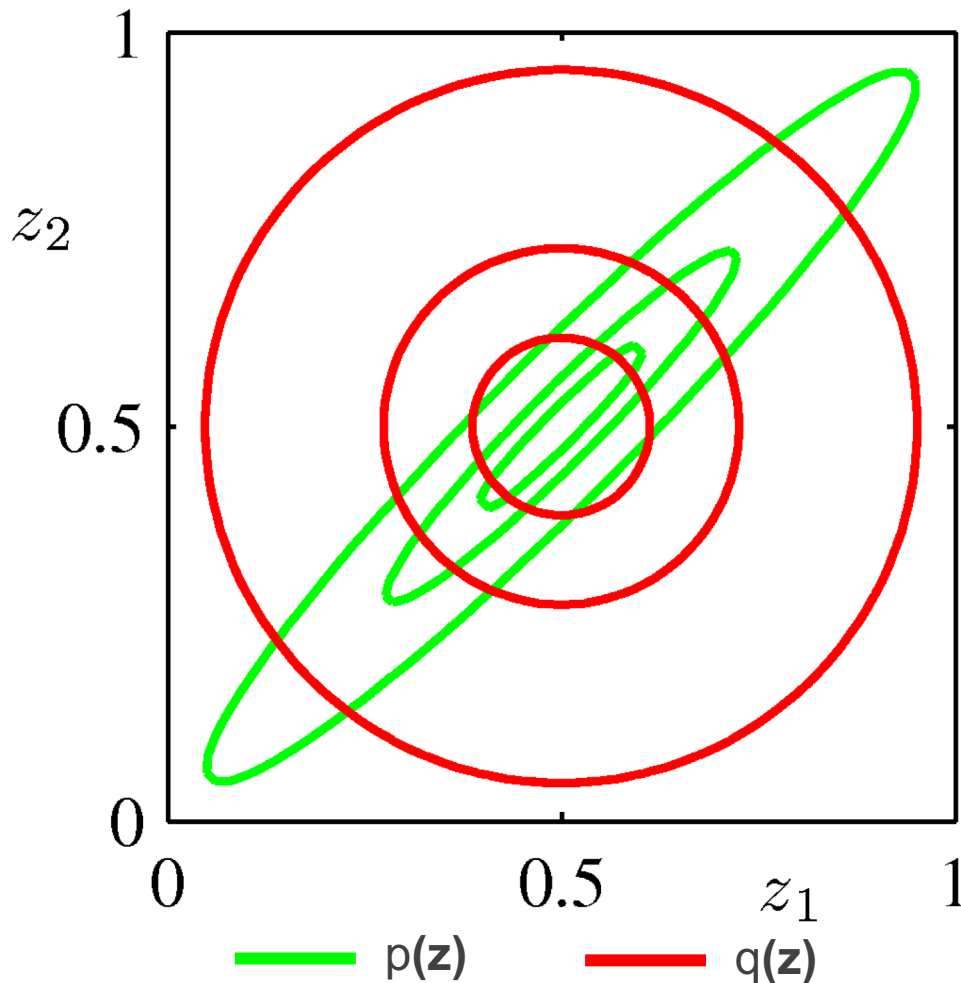
Contours at 1, 2, 3 standard deviations

- Variance controlled by the direction of smallest variance
- Variance on orthogonal direction significantly underestimated
- Factorized variational $\rightarrow$ too compact posteriors

$$\text{KL}(q(\mathbf{z}) \| p(\mathbf{z})) = \mathbb{E}_{q(\mathbf{z})} \left[\log \frac{q(\mathbf{z})}{p(\mathbf{z})} \right]$$

- Solution will avoid regions where $p(\mathbf{z})$ is small

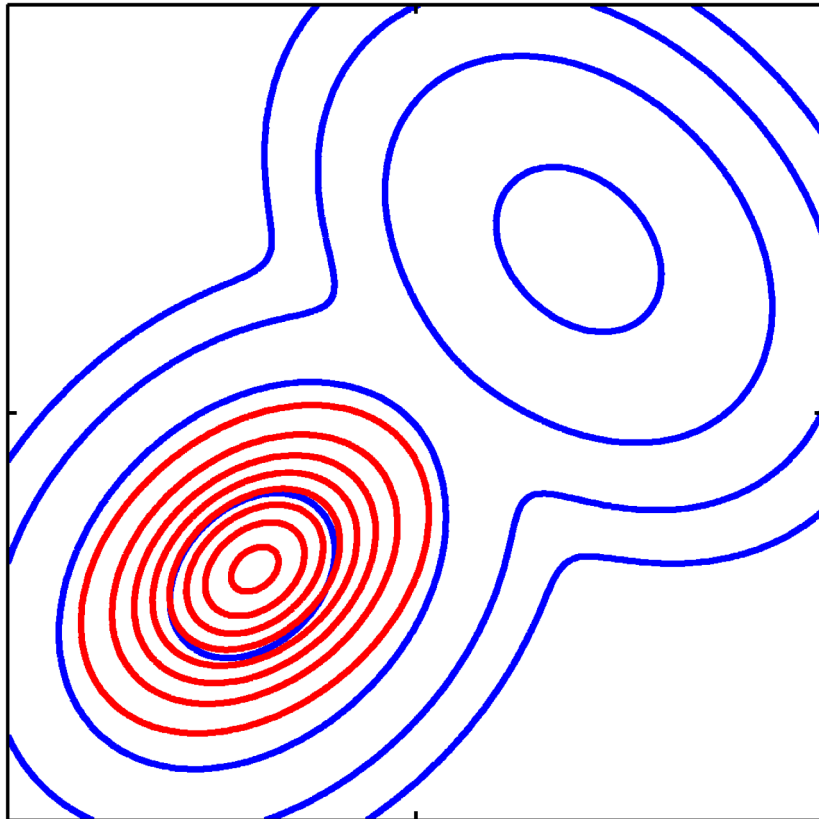
Properties of Factorised Distributions (3)



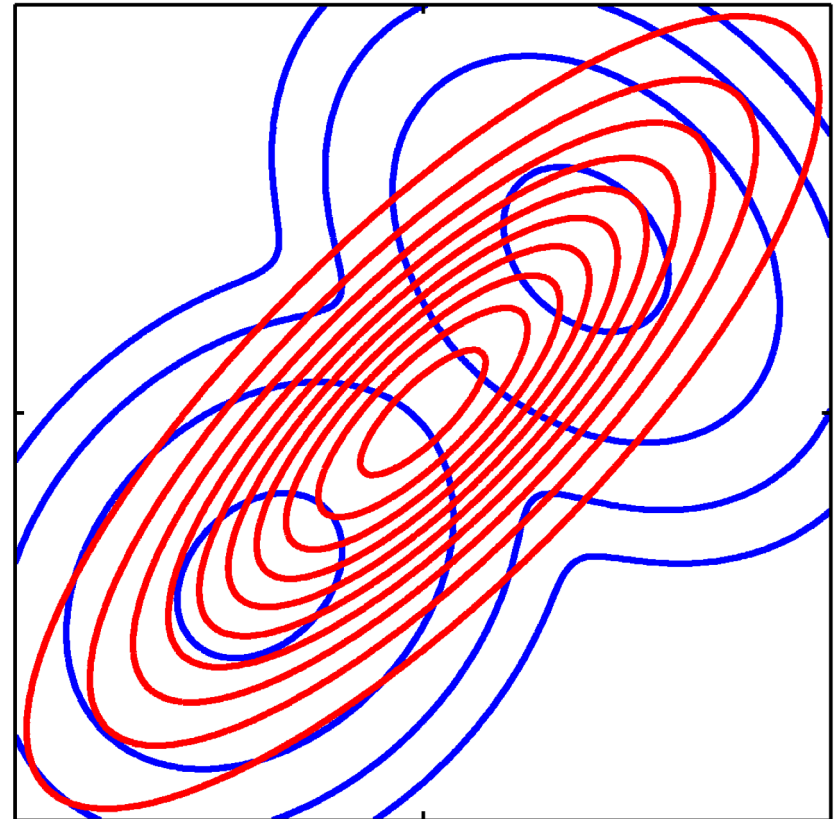
- Minimizing the reverse $KL(p(\mathbf{z})||q(\mathbf{z}))$
 - Expectation propagation
- Solution corresponds to the marginal distributions
- Correct means
- Significant mass in regions of low $p(\mathbf{z})$

Properties of Factorised Distributions (4)

Optimisation of $KL(q||p)$



Optimisation of $KL(p||q)$



— True multimodal posterior $p(\mathbf{z})$
— Unimodal $q(\mathbf{z})$

— True multimodal posterior $p(\mathbf{z})$
— Unimodal $q(\mathbf{z})$